



AI-Powered Cognitive Care & Memory Companion Platform

Culturally Attuned Dementia & Mild Cognitive Impairment (MCI) Support for North-Eastern India

COMPETITION

Smart India Hackathon 2026

Problem Statement: SIH26003

NODAL MINISTRY

Ministry of DoNER

Govt. of India North East Focus

DEVELOPMENT TEAM

Team Design Divas

AI, Web, PWA & Clinical Telemetry

🌟 CORE ARCHITECTURAL PILLARS

- **5 Culturally Rooted Cognitive Games:** NER heritage produce, instruments, and family recall.
- **Multimodal Voice & WhatsApp Bot:** EN/Hi live speech synthesis & automated WhatsApp check-ins.
- **100% Offline-First PWA:** Workbox Service Worker + IndexedDB + Biometric WebAuthn login.
- **Caregiver Command Center:** Real-time Red Flag alerts, ML cognitive scoring & DPDP 2023 compliance.

SIH26003: Cognitive Care in North-Eastern India

Empowering seniors with Alzheimer's, Dementia & MCI through accessible digital intervention

 **THE REGIONAL HEALTHCARE CHALLENGE:** Dementia and MCI are critically underdiagnosed in North-Eastern India due to geographic isolation, scarce specialist access, low digital literacy, and intermittent rural connectivity. Seniors need non-stigmatizing daily cognitive stimulation and caregiver medication monitoring without technical friction.

Core Requirements (a – d)

- **(a) Culturally Attuned Games:** Familiar Indian & NER regional contexts to maximize patient engagement.
- **(b) AI/ML Adaptive Difficulty:** Dynamically tune game challenge based on round accuracy and cognitive history.
- **(c) Cross-Device & Offline Access:** Reliable operation in remote hilly areas with zero connectivity.
- **(d) Multilingual Voice Support:** Seamless regional language interaction (Assamese, Hindi, NE dialects).

Core Requirements (e – h)

- **(e) Longitudinal Telemetry:** Precise clinical tracking of memory scores, reaction times, and cognitive trends.
- **(f) Real-Time Red-Flag Alerts:** Instant notifications to caregivers for missed routines & medication lapses.
- **(g) Low-Barrier Interaction:** Hands-free voice guidance, WhatsApp chatbot, and oversized touch targets.
- **(h) DPDP Act 2023 Compliance:** Explicit photo consent, AES-256 encryption, 30-day erasure guarantees.

The Three Barriers Framework

How Smriti's architecture systematically eliminates structural elderly healthcare hurdles

BARRIER 1

Digital Literacy

Elderly Seniors & Dementia

The Challenge: Password fatigue, complex hierarchies, and fear of navigating intricate smartphone apps.

Smriti Solution:

- Voice-first auto read-aloud on every screen
- WebAuthn 1-tap biometrics & 4-digit PIN
- WhatsApp Companion on familiar chat app
- 56px tactile touch targets & warm contrast

BARRIER 2

Connectivity & Power

Intermittent 2G/3G in NER

The Challenge: Frequent network blackouts in rural hills; seniors lose progress if disconnected.

Smriti Solution:

- Standalone PWA app shell via Workbox
- IndexedDB caches patient profiles & routines
- Background sync auto-pushes offline data
- Ultra-fast, lightweight sub-second bundle

BARRIER 3

Language & Culture

Western Tests Alienating Patients

The Challenge: Standard clinical tests use foreign concepts, causing patient anxiety and false declines.

Smriti Solution:

- NER Themes: Kaji Nemu, Bihu Dhol, Tulsi puja
- Multilingual UI: English, Hindi & Assamese
- Bhashini ULCA pipeline for NE dialects
- Family photo reminiscence therapy

Full-Stack Architecture Specification

Production-grade stack engineered for high responsiveness, resilience, and clinical reliability

Layer / Subsystem	Technologies & Libraries	Key Capabilities & Implementation Role
Frontend Client	React 19, Vite 8, React Router v7, Tailwind CSS v4	Sub-second rendering, tactile senior components, accessible 56px touch targets, warm theme.
Backend Server	Node.js, Express.js (v5), Node-Cron, Webhooks	REST API, WebAuthn challenge verification, JWT middleware, automated scheduled cron triggers.
Database & Storage	MongoDB Atlas, Mongoose v9, IndexedDB (idb)	Document storage for Patient profiles, Reminders, Photos, and Game telemetry with offline sync.
Conversational AI	Google Gemini AI SDK (@google/genai)	Grounded in live MongoDB patient profile & routine data for empathetic, hallucination-free support.
ML Telemetry Engine	Python FastAPIs on Render (Live ML Service)	Endpoints /get_next_difficulty (adaptive starting level) & /calculate_health_score.
Voice & Audio	Web Speech Synthesis & Recognition API	Hands-free microphone voice queries and spoken audio read-aloud (Listen buttons) in EN/HI.
Messaging Gateway	Meta WhatsApp Business Cloud API (wa.me)	Consolidated daily routine alerts, automated "DONE" response parsing & bi-directional bot check-ins.
Offline PWA	vite-plugin-pwa, Workbox Service Worker	Complete standalone app caching, background action sync queue, offline routine checklist.
Auth & Governance	WebAuthn (FIDO2), JWT (365-day), DPDP Act 2023	Biometric 1-tap login, explicit photo upload consent, AES-256 encryption, Grievance Officer.

Senior-Centric Patient Experience

Empowering independence through accessible design, time-aware horizons, and multi-modal guidance

ENGAGEMENT

Games-First Dashboard

Prominently features cognitive games right on the home screen to encourage daily brain stimulation before checking routine tasks.

ADHERENCE

Time-Aware Horizons

Smart time segmentation (Morning, Afternoon, Evening, Night) automatically surfaces active medications and hydration goals.

ACCESSIBILITY

Universal Audio (Listen)

Every card and reminder features a high-contrast [ Listen] button powered by Web Speech API for seniors with low reading literacy.

CONVERSATIONAL

Grounded Gemini AI

Floating sparkle assistant with voice STT/TTS, answering health and schedule questions grounded dynamically in patient data.

COMMUNICATION

1-Tap WhatsApp Bridge

Dedicated green launcher connecting seniors to the Smriti WhatsApp companion bot without searching phone contact lists.

SECURITY

Frictionless Auth

1-Tap hardware biometric fingerprint/Face ID login or simplified 4-digit PIN pad with persistent 365-day sessions.

The 5 Culturally Attuned Cognitive Games

Culturally familiar Indian & North-Eastern themes designed to maximize reminiscence and clinical telemetry

Game Title	Cognitive Domain	Cultural Context (NER Theme)	Clinical Goal & Mechanics
Market Day Basket (বজাৰৰ পাচি / बाज़ार की टोकरी)	Categorization, Visual Recall & Mental Math	10 North-East produce items (Kaji Nemu, King Chilli, Assam Tea, Bamboo Shoot, Lai Xaak).	Scales item count dynamically. Evaluates visual memory retention and price summation accuracy.
Daily Routine Sequencer (দৈনিক দিনচৰ্চা ক্ৰম)	Executive Function & Logical Step Sequencing	Familiar Indian morning routines (brewing Assam tea, watering sacred Tulsi plant, morning bath).	Measures chronological reasoning, step ordering logic, hesitation time, and error correction rates.
Faces & Family Recall (চেহৰা আৰু পৰিয়াল / चेहरे और परिवार)	Reminiscence Therapy & Facial Memory	Real family member portraits, generational relationships (Grandson, Daughter, Spouse).	Reinforces facial recognition, reduces caregiver alienation, and tracks associative recall latency.
Sound & Rhythm Match (শব্দ আৰু ছন্দ / ध्वनि और लय)	Auditory Perception & Pattern Memory	Authentic regional instruments (Bihu Dhol, Pepa flute, Temple Bell, Shankha, Mridangam).	Tests auditory sequence repetition, musical cadence recognition, and rhythmic reaction timing.
Odd One Out (সুকীয়া বস্তু বাছনি / अलग पहचानें)	Semantic Discrimination & Category Boundaries	Indigenous flora, fauna, traditional kitchen tools (Japi, Xorai), and cultural festivals.	Detects semantic anomaly discrimination, visual processing latency, and category boundary confusion.

Clinical Caregiver Command Center

Real-time supervision, compliance tracking, and longitudinal cognitive analytics for clinicians & families

Real-Time Red Flag Alerts

Instant Intervention for Lapsed Medication

- Real-time detection of unacknowledged critical prescriptions.
- 1-Tap direct phone call button to immediately reach the patient.
- Database-persisted dismissal audit log with clinician acknowledgement.

7-Day Adherence Analytics

Deterministic Compliance Charts

- Longitudinal line charts tracking medication & hydration completion.
- 10-slot customizable routine editor with time-of-day scheduling.
- Exportable clinical compliance logs for attending neurologists.

Cognitive Telemetry Dashboard

Longitudinal Session Insights

- Deep analytics tracking reaction speed, hesitation time, and accuracy.
- Game-by-game breakdown across all 5 cognitive domains.
- Weekly Composite Cognitive Health index powered by AI ML API.

Multi-Patient Roster Management

Clinic & Hospital Multi-Tenancy

- Caregivers monitor multiple distinct patients across NER clinics.
- Strict per-caregiver data isolation with role-scoped JWT security.
- Rapid patient profile switching with individual photo galleries.

WhatsApp Companion Bot (Meta Cloud API)

Meeting elderly patients on their most familiar messaging application without requiring app navigation

💬 Automated WhatsApp Bot Pipeline

1. **Node-Cron Daily Trigger:** Evaluates patient routine times (09:00, 14:00, 20:00).
2. **Meta Cloud API Outbound Dispatch:** Sends consolidated WhatsApp reminder to patient phone.
3. **1-Tap Link (wa.me/15556680031):** Patient receives alert or initiates conversation directly.
4. **Automated "DONE" Acknowledgment:** Replying "Done", "Ho gaya" or "হ'ল" ticks routine in DB.
5. **Multi-Turn Gemini AI Webhook:** Natural conversational Q&A grounded in patient context.

✓ Zero-Friction Routine Check-Ins

Consolidated daily routine schedules sent right to WhatsApp. Elderly users do not need to remember URLs or passwords.

✓ Bidirectional Synchronization

When a senior marks a medicine done via WhatsApp, the Caregiver Command Center updates instantaneously in real time.

✓ Empathetic Conversational Care

Powered by Gemini AI webhook integration, providing warm reassurance, memory prompts, and calming guidance.

Dedicated AI & Machine Learning Architecture

Rigorous, honest, and production-integrated AI systems for personalization and cognitive scoring

CONVERSATIONAL LLM

Google Gemini AI (Live)

- In-app conversational companion powered by official @google/genai SDK.
- **Real-time Grounding:** Injects MongoDB patient records (active medications, daily routines, family relations).
- **Dementia Guardrails:** Calming tone, simple sentences, strictly avoids medical misinformation.

ADAPTIVE SCORING API

Dedicated ML Engine on Render (Live)

- Hosted at <https://dementia-ai-engine.onrender.com> (Live Integration).
- POST /get_next_difficulty: Evaluates past session history to calibrate adaptive starting difficulty.
- POST /calculate_health_score: Computes 0–100 composite cognitive health index from telemetry.

REAL-TIME SCALING

Within-Session Accuracy Adaptation

- Round-to-round dynamic difficulty adjustment directly during gameplay.
- Low accuracy (<60%) triggers automatic deceleration and prompt hints.
- High accuracy (>85%) progressively introduces more complex item sets.
- **Offline Fallback:** Deterministic heuristic calibration if API is disconnected.

CLINICAL RIGOR

Analytics Transparency & Classification

- **7-Day Adherence Line Chart:** Explicitly classified as deterministic MongoDB calculation from real daily routine completion logs.
- **Honest Separation:** Clear demarcation between deterministic adherence telemetry and predictive/adaptive ML models.

Multilingual Access & Bhashini Integration

Bridging linguistic diversity across North-East India through regional speech models

● Current Production State (Live)

✓ Web Speech API (Voice Synthesis & Recognition):

Natural speech read-aloud and hands-free microphone input for English and Hindi (hi-IN) across all dashboards and games.

✓ Assamese (অসমীয়া) UI Localization:

Complete internationalization covering patient navigation, reminders, routines, and cognitive game prompts.

✓ Tactile Visual Signifiers:

Iconography, high-contrast color badges, and phonetic labels enabling non-readers to operate the app effortlessly.

🚀 Bhashini Integration (In Progress)

⚡ Digital India Bhashini ULCA Pipeline:

Architecture designed to connect with Govt. of India Bhashini microservices for North-Eastern languages.

⚡ Targeted NER Language Models:

Integration underway for native Assamese, Khasi, Mizo, and Bengali ASR (speech-to-text) and TTS (text-to-speech).

⚡ Modular Microservice Adapter:

Backend API wrapper ready to route voice requests to Bhashini endpoints with offline fallback caching.

Offline-First PWA & DPDP Act 2023 Compliance

Engineered for rural network resilience and full compliance with Indian data protection laws

Offline-First PWA Architecture

- **Workbox Service Worker:** Caches complete app bundle for standalone installation on Android, iOS, and PC.
- **IndexedDB Local Storage (idb):** Caches senior profiles, family reminiscence photos, and 10 daily routines on device.
- **Resilient Action Sync Queue:** Routine acknowledgements and game scores logged offline auto-sync when internet resumes.
- **Zero App Crash Policy:** Graceful offline fallbacks prevent blank screens or connection error dialogs.

DPDP Act 2023 Compliance

- **Mandatory Photo Consent Checkbox:** Explicit caregiver agreement under DPDP 2023 required before uploading any senior portrait.
- **Public Legal Policy Route (/privacy-policy):** Transparent disclosure of data types, processing purposes, and security safeguards.
- **AES-256 Storage & Scoped JWTs:** Multi-tenant data isolation; biometrics remain on-device (FIDO2 public key standard).
- **30-Day Erasure & Grievance Redressal:** Complete data erasure guarantee upon request with appointed Grievance Officer.

SIH26003 Requirement Coverage Matrix

Comprehensive mapping of Smart India Hackathon problem statement specifications against Smriti delivery

Req	SIH26003 Specification	Smriti Implementation Details	Status
(a)	Culturally Attuned Cognitive Games	5 games themed around NER produce, instruments, routines, and family reminiscence.	DONE (100%)
(b)	AI/ML Adaptive Difficulty Scaling	Live Render ML API (/get_next_difficulty) + real-time within-session scaling.	DONE (100%)
(c)	Cross-Device & Offline PWA Access	Workbox service worker, IndexedDB local caching, and background sync queue.	DONE (100%)
(d)	Multilingual Voice & Text Support	Web Speech API (EN/HI live), Assamese UI live; Bhashini NER voice in progress.	PARTIAL (80%)
(e)	Longitudinal Cognitive Telemetry	Reaction speed, accuracy metrics, session history, and ML health score index.	DONE (100%)
(f)	Real-Time Red-Flag Caregiver Alerts	Missed routine alerts, 1-tap patient calling button, and database dismissal audit.	DONE (100%)
(g)	Low-Barrier Senior Interaction	Voice-first read aloud, Meta WhatsApp bot, 56px touch targets, WebAuthn biometrics.	DONE (100%)
(h)	Data Privacy & DPDP Compliance	Mandatory consent checkboxes, AES-256 encryption, 30-day erasure, Grievance Officer.	DONE (100%)

Scalability & Future Development Roadmap

Expanding Smriti from a digital companion into a nationwide community healthcare infrastructure

PHASE 1 (NEXT 30D)

Full Bhashini Voice

Deploy native Assamese, Khasi, Mizo, and Bengali ASR/TTS models via Bhashini microservices for zero-barrier regional voice interactions.

PHASE 2 (Q2 2026)

Tier 2 IVR / SMS

Automated interactive voice response phone calls and SMS reminders for non-smartphone feature phones in remote hilly terrains.

PHASE 3 (Q3 2026)

Tier 3 ASHA Kiosk

Community tablet interface empowering ASHA healthcare workers to administer standardized weekly cognitive assessments in villages.

PHASE 4 (Q4 2026)

Predictive Decline AI

Multi-month longitudinal machine learning models predicting clinical decline trajectories and alerting clinicians prior to stage transitions.



Restoring Dignity • Preserving Memories • Empowering Caregivers

Thank you to the Judges, Ministry of DoNER & Smart India Hackathon 2026

Team Design Divas

- **Competition:** Smart India Hackathon 2026 Finalist Team
- **Problem Statement:** SIH26003 (Ministry of DoNER)
- **Domain:** AI-Augmented Healthcare & Cognitive Care
- **Contact:** gopalrajlohiya2@gmail.com

Live Repository & Deployments

- **Live Frontend:** Deployed on Vercel SPA Engine
- **Live Backend:** Node.js / MongoDB Atlas on Render
- **Live ML Engine:** <https://dementia-ai-engine.onrender.com>
- **WhatsApp Bot:** <https://wa.me/15556680031>